

PROJECT DESIGN PHASE – II

Data Flow Diagram & User Stories

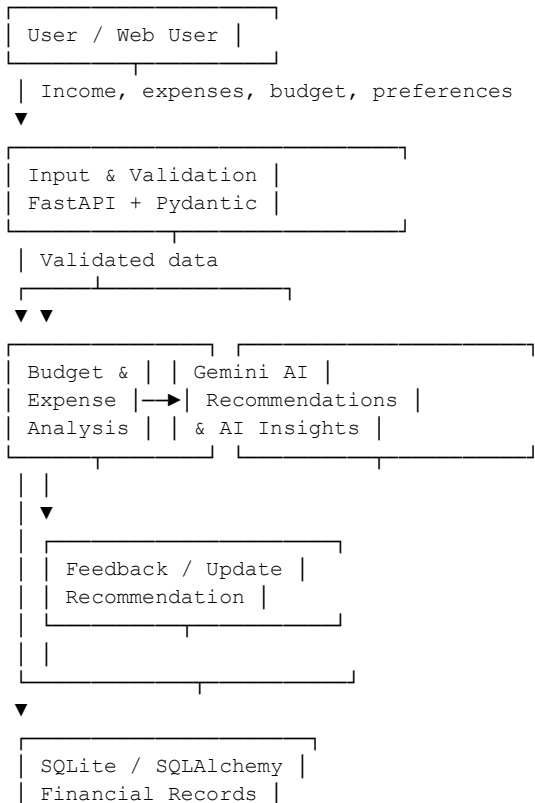
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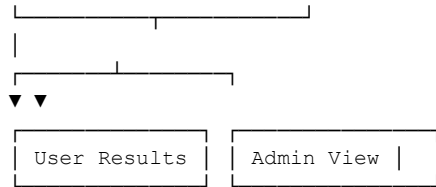
1. Project & Team Details

Team ID	SWTID-2026-3350
Project Name	PocketSmart AI – Your Smart Budget & Recommendation Assistant
Team Size	4
Team Members	Esther Ammo M , Epsipa R , Eswari M , Fayisa Fathima A

2. Data Flow Diagram (DFD)

The DFD represents the main information flow in PocketSmart AI: user financial details are entered and validated, processed for budget and expense analysis, sent to the Gemini AI layer for personalized recommendations, stored in SQLite, and later used for feedback-based recommendation updates and administrator viewing.





3. DFD Components

DFD Element	Description
User	Provides income, budget amount, expense details, spending categories and preferences; can submit feedback.
User Input & Validation	Receives form data and validates required financial fields before processing.
Budget & Expense Analysis	Processes income, budget and expense data to calculate spending totals, remaining budget and financial summaries.
Gemini AI	Generates personalized budgeting insights and recommendations and supports feedback-based recommendation updates.
Feedback & Recommendation Update	Processes user feedback or updated financial information and produces an updated recommendation.
SQLite / SQLAlchemy	Stores user information, budgets, expenses and recommendation records.
Admin View	Allows authorized administrators to view registered users and relevant budget, expense and recommendation information.

4. User Stories

The following user stories use the supplied template structure: User Type, Functional Requirement (Epic), User Story Number, User Story / Task, Acceptance Criteria, Priority and Release.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	Profile & Financial Input	USN-1	As a user, I can enter my income, budget amount, expenses, expense categories and spending preferences.	All required financial inputs are captured and validated.	High	Sprint-1
Customer (Web User)	Budget & Expense Analysis	USN-2	As a user, I can view a summary of my income, spending, budget and remaining amount.	Budget and expense values are processed and a clear financial summary is displayed.	High	Sprint-2

Customer (Web User)	AI Recommendation Generation	USN-3	As a user, I can receive personalized budgeting and spending recommendations based on my financial information.	A relevant AI-generated recommendation is displayed with the financial summary.	High	Sprint-2
Customer (Web User)	Financial Data Storage	USN-4	As a user, I can have my financial details and generated recommendations stored for later access.	User financial records and recommendation information are stored in SQLite.	High	Sprint-2
Customer (Web User)	Feedback	USN-5	As a user, I can submit feedback or updated information such as a changed budget, expense or spending preference.	Feedback or updated information is accepted and linked to the relevant user context.	High	Sprint-3
Customer (Web User)	Recommendation Update	USN-6	As a user, I can receive an updated recommendation based on my feedback or latest financial information.	Previous context and new feedback are processed and an updated recommendation is displayed.	High	Sprint-3
Customer (Web User)	Recommendation History	USN-7	As a user, I can preserve previous recommendation information while an updated version is stored separately.	Original and updated recommendation records remain available.	Medium	Sprint-3
Administrator	Admin Dashboard	USN-8	As an administrator, I can view registered users and their relevant financial and recommendation records.	User and financial information is shown in a structured administrative view according to access controls.	Medium	Sprint-3
Administrator	Recommendation Monitoring	USN-9	As an administrator, I can compare previous and updated recommendation information for a user.	Relevant recommendation versions and associated information are displayed clearly.	Medium	Sprint-3

Developer	FastAPI Backend	USN-10	As a developer, I can implement routes for financial input, budget analysis, recommendation generation, feedback and admin view.	Required FastAPI routes process requests and return correct responses or templates.	High	Sprint-3
Developer	Frontend	USN-11	As a developer, I can provide responsive Jinja2 pages for financial input, results, recommendations and administration.	Required pages render correctly and display dynamic data.	High	Sprint-4
Developer	Deployment & Testing	USN-12	As a developer, I can run and verify PocketSmart AI locally using Uvicorn.	The application loads locally, core workflows can be tested, and API documentation is accessible.	High	Sprint-4

5. User Story Acceptance Summary

A user story is considered complete when its acceptance criteria are satisfied, the relevant PocketSmart AI functionality works as intended, and the feature is verified during the assigned sprint.

6. Main PocketSmart AI Functional Epics

- Profile & Financial Input
- Budget & Expense Analysis
- Personalized AI Recommendation Generation
- Database & Financial Data Storage
- Feedback-Based Recommendation Update
- FastAPI Routing & Backend
- Frontend / Jinja2 Interface
- Administrator Dashboard
- Deployment & Testing

